

# hw3

November 2, 2025

```
[1]: ## IMPORTANT: On Colab, we expect your homework to be in the cs189 folder  
## Please contact staff if you encounter any problems with installing  
↳dependencies  
import sys  
IS_COLAB = 'google.colab' in sys.modules  
if IS_COLAB:  
    from google.colab import drive  
    drive.mount('/content/drive')  
    %cd /content/drive/MyDrive/cs189/hw/hw3  
    %pip install -r ./requirements.txt  
    !pip install -U kaleido plotly  
    import kaleido  
    kaleido.get_chrome_sync()  
  
import plotly.io as pio  
pio.renderers.default = pio.renderers.default + "+png"
```

```
Mounted at /content/drive  
/content/drive/MyDrive/cs189/hw/hw3  
Collecting otter_grader (from -r ./requirements.txt (line 1))  
  Cloning https://github.com/BerkeleyML/otter-grader-189.git (to revision  
master) to /tmp/pip-install-swp7_uzu/otter-  
grader_63f43b3c45394a96b00cf6b57215498c  
  Running command git clone --filter=blob:none --quiet  
https://github.com/BerkeleyML/otter-grader-189.git /tmp/pip-install-  
swp7_uzu/otter-grader_63f43b3c45394a96b00cf6b57215498c  
  Resolved https://github.com/BerkeleyML/otter-grader-189.git to commit  
35e43fdf64a55f4b9222d7b531890fb7d5cb498c  
  Installing build dependencies ... done  
  Getting requirements to build wheel ... done  
  Preparing metadata (pyproject.toml) ... done  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages  
(from -r ./requirements.txt (line 2)) (4.67.1)  
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-  
packages (from -r ./requirements.txt (line 3)) (3.10.0)  
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages  
(from -r ./requirements.txt (line 4)) (5.24.1)  
Requirement already satisfied: numpy>=1.21.0 in /usr/local/lib/python3.12/dist-
```

```

packages (from -r ./requirements.txt (line 5)) (2.0.2)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 6)) (2.2.2)
Requirement already satisfied: scikit-learn>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 7))
(1.6.1)
Requirement already satisfied: pytest>=6.0.0 in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 8)) (8.4.2)
Collecting jupyter>=1.0.0 (from -r ./requirements.txt (line 9))
  Downloading jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: tabulate in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 10)) (0.9.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 11)) (2.8.0+cu126)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 12)) (0.23.0+cu126)
Collecting tiktoken==0.11.0 (from -r ./requirements.txt (line 13))
  Downloading tiktoken-0.11.0-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.7 kB)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 14)) (4.0.0)
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 16)) (5.49.1)
Requirement already satisfied: huggingface_hub[cli,torch] in
/usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 15))
(0.36.0)
Requirement already satisfied: regex>=2022.1.18 in
/usr/local/lib/python3.12/dist-packages (from tiktoken==0.11.0->-r
./requirements.txt (line 13)) (2024.11.6)
Requirement already satisfied: requests>=2.26.0 in
/usr/local/lib/python3.12/dist-packages (from tiktoken==0.11.0->-r
./requirements.txt (line 13)) (2.32.4)
Requirement already satisfied: click<9.0.0,>=8.1.7 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (8.3.0)
Requirement already satisfied: dill>=0.3.0 in /usr/local/lib/python3.12/dist-
packages (from otter_grader->-r ./requirements.txt (line 1)) (0.3.8)
Collecting fica>=0.4.1 (from otter_grader->-r ./requirements.txt (line 1))
  Downloading fica-0.4.1-py3-none-any.whl.metadata (2.0 kB)
Collecting ipylab<2.0.0,>=1.0.0 (from otter_grader->-r ./requirements.txt (line
1))
  Downloading ipylab-1.1.0-py3-none-any.whl.metadata (6.7 kB)
Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-
packages (from otter_grader->-r ./requirements.txt (line 1)) (7.34.0)
Collecting ipywidgets<9.0.0,>=8.1.5 (from otter_grader->-r ./requirements.txt
(line 1))
  Downloading ipywidgets-8.1.8-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: jinja2<4.0,>=3.1 in

```

```

/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: jupyter<2.0.0,>=1.16.4 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (1.17.3)
Requirement already satisfied: nbconvert>=6.0.0 in
/usr/local/lib/python3.12/dist-packages (from nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (7.16.6)
Requirement already satisfied: nbformat>=5.0.0 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (5.10.4)
Collecting python-on-whales<1.0.0,>=0.72.0 (from otter_grader->-r
./requirements.txt (line 1))
  Downloading python_on_whales-0.79.0-py3-none-any.whl.metadata (18 kB)
Requirement already satisfied: pyyaml<7,>=6 in /usr/local/lib/python3.12/dist-
packages (from otter_grader->-r ./requirements.txt (line 1)) (6.0.3)
Collecting wrapt<2.0.0,>=1.16.0 (from otter_grader->-r ./requirements.txt (line
1))
  Downloading wrapt-1.17.3-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (6.4
kB)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (1.3.3)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->-r ./requirements.txt (line 3)) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (1.4.9)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->-r ./requirements.txt (line 3)) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt
(line 3)) (2.9.0.post0)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->-r ./requirements.txt
(line 4)) (8.5.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-

```

packages (from pandas>=1.3.0->-r ./requirements.txt (line 6)) (2025.2)  
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->-r ./requirements.txt (line 6)) (2025.2)  
 Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (1.16.3)  
 Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (1.5.2)  
 Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (3.6.0)  
 Requirement already satisfied: iniconfig>=1 in /usr/local/lib/python3.12/dist-packages (from pytest>=6.0.0->-r ./requirements.txt (line 8)) (2.3.0)  
 Requirement already satisfied: pluggy<2,>=1.5 in /usr/local/lib/python3.12/dist-packages (from pytest>=6.0.0->-r ./requirements.txt (line 8)) (1.6.0)  
 Requirement already satisfied: pygments>=2.7.2 in /usr/local/lib/python3.12/dist-packages (from pytest>=6.0.0->-r ./requirements.txt (line 8)) (2.19.2)  
 Requirement already satisfied: notebook in /usr/local/lib/python3.12/dist-packages (from jupyter>=1.0.0->-r ./requirements.txt (line 9)) (6.5.7)  
 Requirement already satisfied: jupyter-console in /usr/local/lib/python3.12/dist-packages (from jupyter>=1.0.0->-r ./requirements.txt (line 9)) (6.6.3)  
 Requirement already satisfied: ipykernel in /usr/local/lib/python3.12/dist-packages (from jupyter>=1.0.0->-r ./requirements.txt (line 9)) (6.17.1)  
 Collecting jupyterlab (from jupyter>=1.0.0->-r ./requirements.txt (line 9))  
   Downloading jupyterlab-4.4.10-py3-none-any.whl.metadata (16 kB)  
 Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (3.20.0)  
 Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (4.15.0)  
 Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (75.2.0)  
 Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (1.13.3)  
 Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (3.5)  
 Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (2025.3.0)  
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (12.6.77)  
 Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (12.6.77)  
 Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line 11)) (12.6.80)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (9.10.2.21)

Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (12.6.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (11.3.0.4)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (10.3.7.77)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (11.7.1.2)

Requirement already satisfied: nvidia-cuspars-cu12==12.5.4.2 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (12.5.4.2)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (12.5.4.2)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (2.27.3)

Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (12.6.77)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (12.6.85)

Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in  
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line  
 11)) (1.11.1.6)

Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-  
 packages (from torch->-r ./requirements.txt (line 11)) (3.4.0)

Requirement already satisfied: pyarrow>=15.0.0 in  
 /usr/local/lib/python3.12/dist-packages (from datasets->-r ./requirements.txt  
 (line 14)) (18.1.0)

Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages  
 (from datasets->-r ./requirements.txt (line 14)) (3.6.0)

Requirement already satisfied: multiprocessing<0.70.17 in  
 /usr/local/lib/python3.12/dist-packages (from datasets->-r ./requirements.txt  
 (line 14)) (0.70.16)

Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in  
 /usr/local/lib/python3.12/dist-packages (from huggingface\_hub[cli,torch]->-r  
 ./requirements.txt (line 15)) (1.2.0)

Collecting InquirerPy==0.3.4 (from huggingface\_hub[cli,torch]->-r  
 ./requirements.txt (line 15))

```

    Downloading InquirerPy-0.3.4-py3-none-any.whl.metadata (8.1 kB)
Requirement already satisfied: safetensors[torch] in
/usr/local/lib/python3.12/dist-packages (from huggingface_hub[cli,torch]->-r
./requirements.txt (line 15)) (0.6.2)
Collecting pfzy<0.4.0,>=0.3.1 (from
InquirerPy==0.3.4->huggingface_hub[cli,torch]->-r ./requirements.txt (line 15))
    Downloading pfzy-0.3.4-py3-none-any.whl.metadata (4.9 kB)
Requirement already satisfied: prompt-toolkit<4.0.0,>=3.0.1 in
/usr/local/lib/python3.12/dist-packages (from
InquirerPy==0.3.4->huggingface_hub[cli,torch]->-r ./requirements.txt (line 15))
(3.0.52)
Requirement already satisfied: aiofiles<25.0,>=22.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (4.11.0)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.12/dist-
packages (from gradio->-r ./requirements.txt (line 16)) (1.1.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.120.1)
Requirement already satisfied: ffmpeg in /usr/local/lib/python3.12/dist-packages
(from gradio->-r ./requirements.txt (line 16)) (0.6.4)
Requirement already satisfied: gradio-client==1.13.3 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (1.13.3)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.12/dist-
packages (from gradio->-r ./requirements.txt (line 16)) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.28.1)
Requirement already satisfied: markupsafe<4.0,>=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (3.0.3)
Requirement already satisfied: orjson~=3.0 in /usr/local/lib/python3.12/dist-
packages (from gradio->-r ./requirements.txt (line 16)) (3.11.4)
Requirement already satisfied: pydantic<2.12,>=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (2.11.10)
Requirement already satisfied: pydub in /usr/local/lib/python3.12/dist-packages
(from gradio->-r ./requirements.txt (line 16)) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.0.20)
Requirement already satisfied: ruff>=0.9.3 in /usr/local/lib/python3.12/dist-
packages (from gradio->-r ./requirements.txt (line 16)) (0.14.2)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in

```

```

/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.49.1)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.20.0)
Requirement already satisfied: uvicorn>=0.14.0 in
/usr/local/lib/python3.12/dist-packages (from gradio->-r ./requirements.txt
(line 16)) (0.38.0)
Requirement already satisfied: websockets<16.0,>=13.0 in
/usr/local/lib/python3.12/dist-packages (from gradio-client==1.13.3->gradio->-r
./requirements.txt (line 16)) (15.0.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-
packages (from anyio<5.0,>=3.0->gradio->-r ./requirements.txt (line 16)) (3.11)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-
packages (from anyio<5.0,>=3.0->gradio->-r ./requirements.txt (line 16)) (1.3.1)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2->gradio->-r
./requirements.txt (line 16)) (0.0.3)
Requirement already satisfied: docutils in /usr/local/lib/python3.12/dist-
packages (from fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1))
(0.21.2)
Requirement already satisfied: sphinx in /usr/local/lib/python3.12/dist-packages
(from fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (8.2.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r ./requirements.txt (line 14))
(3.13.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
packages (from httpx<1.0,>=0.24.1->gradio->-r ./requirements.txt (line 16))
(2025.10.5)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-
packages (from httpx<1.0,>=0.24.1->gradio->-r ./requirements.txt (line 16))
(1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio->-r ./requirements.txt
(line 16)) (0.16.0)
Collecting comm>=0.1.3 (from ipywidgets<9.0.0,>=8.1.5->otter_grader->-r
./requirements.txt (line 1))
  Downloading comm-0.2.3-py3-none-any.whl.metadata (3.7 kB)

```

Requirement already satisfied: traitlets>=4.3.1 in  
 /usr/local/lib/python3.12/dist-packages (from  
 ipywidgets<9.0.0,>=8.1.5->otter\_grader->-r ./requirements.txt (line 1)) (5.7.1)  
 Collecting widgetsnbextension~=4.0.14 (from  
 ipywidgets<9.0.0,>=8.1.5->otter\_grader->-r ./requirements.txt (line 1))  
 Downloading widgetsnbextension-4.0.15-py3-none-any.whl.metadata (1.6 kB)  
 Requirement already satisfied: jupyterlab\_widgets~=3.0.15 in  
 /usr/local/lib/python3.12/dist-packages (from  
 ipywidgets<9.0.0,>=8.1.5->otter\_grader->-r ./requirements.txt (line 1)) (3.0.15)  
 Collecting jedi>=0.16 (from ipython->otter\_grader->-r ./requirements.txt (line  
 1))  
 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)  
 Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-  
 packages (from ipython->otter\_grader->-r ./requirements.txt (line 1)) (4.4.2)  
 Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-  
 packages (from ipython->otter\_grader->-r ./requirements.txt (line 1)) (0.7.5)  
 Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-  
 packages (from ipython->otter\_grader->-r ./requirements.txt (line 1)) (0.2.0)  
 Requirement already satisfied: matplotlib-inline in  
 /usr/local/lib/python3.12/dist-packages (from ipython->otter\_grader->-r  
 ./requirements.txt (line 1)) (0.2.1)  
 Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-  
 packages (from ipython->otter\_grader->-r ./requirements.txt (line 1)) (4.9.0)  
 Requirement already satisfied: markdown-it-py>=1.0 in  
 /usr/local/lib/python3.12/dist-packages (from  
 jupyter<2.0.0,>=1.16.4->otter\_grader->-r ./requirements.txt (line 1)) (4.0.0)  
 Requirement already satisfied: mdit-py-plugins in  
 /usr/local/lib/python3.12/dist-packages (from  
 jupyter<2.0.0,>=1.16.4->otter\_grader->-r ./requirements.txt (line 1)) (0.5.0)  
 Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-  
 packages (from nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys\_platform !=  
 "emscripten" and sys\_platform != "wasi"->otter\_grader->-r ./requirements.txt  
 (line 1)) (4.13.5)  
 Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-  
 packages (from bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;  
 sys\_platform != "emscripten" and sys\_platform != "wasi"->otter\_grader->-r  
 ./requirements.txt (line 1)) (6.3.0)  
 Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-  
 packages (from nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys\_platform !=  
 "emscripten" and sys\_platform != "wasi"->otter\_grader->-r ./requirements.txt  
 (line 1)) (0.7.1)  
 Requirement already satisfied: jupyter-core>=4.7 in  
 /usr/local/lib/python3.12/dist-packages (from  
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys\_platform != "emscripten" and  
 sys\_platform != "wasi"->otter\_grader->-r ./requirements.txt (line 1)) (5.9.1)  
 Requirement already satisfied: jupyterlab-pygments in  
 /usr/local/lib/python3.12/dist-packages (from  
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys\_platform != "emscripten" and



```

sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in
/usr/local/lib/python3.12/dist-packages (from
nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (3.1.4)
Requirement already satisfied: nbclient>=0.5.0 in
/usr/local/lib/python3.12/dist-packages (from
nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (0.10.2)
Requirement already satisfied: pandocfilters>=1.4.1 in
/usr/local/lib/python3.12/dist-packages (from
nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (1.5.1)
Collecting playwright (from nbconvert[webpdf]>=6.0.0; sys_platform !=
"emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
(line 1))
  Downloading playwright-1.55.0-py3-none-manylinux1_x86_64.whl.metadata (3.5 kB)
Requirement already satisfied: fastjsonschema>=2.15 in
/usr/local/lib/python3.12/dist-packages (from nbformat>=5.0.0->otter_grader->-r
./requirements.txt (line 1)) (2.21.2)
Requirement already satisfied: jsonschema>=2.6 in
/usr/local/lib/python3.12/dist-packages (from nbformat>=5.0.0->otter_grader->-r
./requirements.txt (line 1)) (4.25.1)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio->-r
./requirements.txt (line 16)) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio->-r
./requirements.txt (line 16)) (2.33.2)
Requirement already satisfied: typing-inspection>=0.4.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio->-r
./requirements.txt (line 16)) (0.4.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib->-r ./requirements.txt (line 3))
(1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests>=2.26.0->tiktoken==0.11.0->-r ./requirements.txt (line 13)) (3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests>=2.26.0->tiktoken==0.11.0->-r ./requirements.txt (line 13)) (2.5.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->-r
./requirements.txt (line 11)) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio->-r
./requirements.txt (line 16)) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-

```

packages (from typer<1.0,>=0.12->gradio->-r ./requirements.txt (line 16))  
(13.9.4)

Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
(1.8.15)

Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (7.4.9)

Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
(1.6.0)

Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (5.9.5)

Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
(26.2.1)

Requirement already satisfied: tornado>=6.1 in /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
(6.5.1)

Collecting async-lru>=1.0.0 (from jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
  Downloading async\_lru-2.0.5-py3-none-any.whl.metadata (4.5 kB)

Collecting jupyter-lsp>=2.0.0 (from jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
  Downloading jupyter\_lsp-2.3.0-py3-none-any.whl.metadata (1.8 kB)

Requirement already satisfied: jupyter-server<3,>=2.4.0 in /usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (2.14.0)

Collecting jupyterlab-server<3,>=2.27.1 (from jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
  Downloading jupyterlab\_server-2.28.0-py3-none-any.whl.metadata (5.9 kB)

Requirement already satisfied: notebook-shim>=0.2 in /usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.2.4)

Requirement already satisfied: argon2-cffi in /usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9))  
(25.1.0)

Requirement already satisfied: ipython-genutils in /usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.2.0)

Requirement already satisfied: Send2Trash>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.8.3)

Requirement already satisfied: terminado>=0.8.3 in /usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.18.1)

Requirement already satisfied: prometheus-client in /usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r

```

./requirements.txt (line 9)) (0.23.1)
Requirement already satisfied: nbclassic>=0.4.7 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.3.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-
packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (6.7.0)
Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (0.4.1)
Requirement already satisfied: yarll<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from
aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets->-r
./requirements.txt (line 14)) (1.22.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
packages (from
bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from
bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform !=
"emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
(line 1)) (1.4.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from
jedi>=0.16->ipython->otter_grader->-r ./requirements.txt (line 1)) (0.8.5)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
/usr/local/lib/python3.12/dist-packages (from
jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1))

```

```

(2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in
/usr/local/lib/python3.12/dist-packages (from
jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1))
(0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-
packages (from jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r
./requirements.txt (line 1)) (0.28.0)
Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-
packages (from jupyter-client>=6.1.12->ipykernel->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.4)
Requirement already satisfied: platformdirs>=2.5 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
core>=4.7->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform !=
"emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
(line 1)) (4.5.0)
Requirement already satisfied: jupyter-events>=0.9.0 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(0.12.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(0.5.3)
Requirement already satisfied: overrides>=5.0 in /usr/local/lib/python3.12/dist-
packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (7.7.0)
Requirement already satisfied: websocket-client>=1.7 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(1.9.0)
Requirement already satisfied: argon2-cffi-bindings in
/usr/local/lib/python3.12/dist-packages (from
argon2-cffi->notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (25.1.0)
Requirement already satisfied: babel>=2.10 in /usr/local/lib/python3.12/dist-
packages (from jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (2.17.0)
Collecting json5>=0.9.0 (from jupyterlab-
server<3,>=2.27.1->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
  Downloading json5-0.12.1-py3-none-any.whl.metadata (36 kB)
Requirement already satisfied: mdurl~>0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=1.0->jupyter-text<2.0.0,>=1.16.4->otter_grader->-r
./requirements.txt (line 1)) (0.1.2)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from
pexpect>4.3->ipython->otter_grader->-r ./requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-

```

```

toolkit<4.0.0,>=3.0.1->InquirerPy==0.3.4->huggingface_hub[cli,torch]->-r
./requirements.txt (line 15)) (0.2.14)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-
packages (from beautifulsoup4->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (2.8)
Collecting pyee<14,>=13 (from playwright->nbconvert[webpdf]>=6.0.0; sys_platform
!= "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
(line 1))
  Downloading pyee-13.0.0-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: greenlet<4.0.0,>=3.1.1 in
/usr/local/lib/python3.12/dist-packages (from
playwright->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (3.2.4)
Requirement already satisfied: sphinxcontrib-applehelp>=1.0.7 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-devhelp>=1.0.6 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-htmlhelp>=2.0.6 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.1.0)
Requirement already satisfied: sphinxcontrib-jsmath>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (1.0.1)
Requirement already satisfied: sphinxcontrib-qthelp>=1.0.6 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-serializinghtml>=1.1.9 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: snowballstemmer>=2.2 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (3.0.1)
Requirement already satisfied: alabaster>=0.7.14 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (1.0.0)
Requirement already satisfied: imagesize>=1.3 in /usr/local/lib/python3.12/dist-
packages (from sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line
1)) (1.4.1)
Requirement already satisfied: roman-numerals-py>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (3.1.0)
Requirement already satisfied: python-json-logger>=2.0.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(4.0.0)

```

Requirement already satisfied: rfc3339-validator in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.1.4)

Requirement already satisfied: rfc3986-validator>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.1.1)

Requirement already satisfied: cffi>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from argon2-cffi-bindings->argon2-cffi->notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (2.0.0)

Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi->notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (2.23)

Requirement already satisfied: fqdn in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.5.1)

Requirement already satisfied: isoduration in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (20.11.0)

Requirement already satisfied: jsonpointer>1.13 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (3.0.0)

Requirement already satisfied: rfc3987-syntax>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.1.0)

Requirement already satisfied: uri-template in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.3.0)

Requirement already satisfied: webcolors>=24.6.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (24.11.1)

Requirement already satisfied: lark>=1.2.2 in /usr/local/lib/python3.12/dist-packages (from rfc3987-syntax>=1.1.0->jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.3.1)

Requirement already satisfied: arrow>=0.15.0 in /usr/local/lib/python3.12/dist-

```

packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.4.0)
Downloading
tiktoken-0.11.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.2
MB)
1.2/1.2 MB
19.4 MB/s eta 0:00:00
Downloading jupyter-1.1.1-py2.py3-none-any.whl (2.7 kB)
Downloading InquirerPy-0.3.4-py3-none-any.whl (67 kB)
67.7/67.7 kB
6.2 MB/s eta 0:00:00
Downloading fica-0.4.1-py3-none-any.whl (13 kB)
Downloading ipylab-1.1.0-py3-none-any.whl (101 kB)
101.6/101.6 kB
9.1 MB/s eta 0:00:00
Downloading ipywidgets-8.1.8-py3-none-any.whl (139 kB)
139.8/139.8 kB
11.5 MB/s eta 0:00:00
Downloading python_on_whales-0.79.0-py3-none-any.whl (118 kB)
118.8/118.8 kB
10.0 MB/s eta 0:00:00
Downloading wrapt-1.17.3-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (88 kB)
88.0/88.0 kB
8.5 MB/s eta 0:00:00
Downloading jupyterlab-4.4.10-py3-none-any.whl (12.3 MB)
12.3/12.3 MB
115.2 MB/s eta 0:00:00
Downloading async_lru-2.0.5-py3-none-any.whl (6.1 kB)
Downloading comm-0.2.3-py3-none-any.whl (7.3 kB)
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
76.9 MB/s eta 0:00:00
Downloading jupyter_lsp-2.3.0-py3-none-any.whl (76 kB)
76.7/76.7 kB
4.2 MB/s eta 0:00:00
Downloading jupyterlab_server-2.28.0-py3-none-any.whl (59 kB)
59.8/59.8 kB
4.2 MB/s eta 0:00:00
Downloading pfzy-0.3.4-py3-none-any.whl (8.5 kB)
Downloading widgetsnbextension-4.0.15-py3-none-any.whl (2.2 MB)
2.2/2.2 MB
64.9 MB/s eta 0:00:00
Downloading playwright-1.55.0-py3-none-manylinux1_x86_64.whl (45.9 MB)
45.9/45.9 MB
15.9 MB/s eta 0:00:00
Downloading json5-0.12.1-py3-none-any.whl (36 kB)

```

```

Downloading pyee-13.0.0-py3-none-any.whl (15 kB)
Building wheels for collected packages: otter_grader
  Building wheel for otter_grader (pyproject.toml) ... done
  Created wheel for otter_grader: filename=otter_grader-6.1.3-py3-none-any.whl
size=142409
sha256=9635982965081b549d1a1cef65a95dc3d2fec63425cc7d1eaff32d612bea158d
  Stored in directory: /tmp/pip-ephem-wheel-cache-
vpjq6hx3/wheels/6c/c4/a1/ed40bf2584b0c3446e20b8a0aea721771c46fe292cffe8e50b
Successfully built otter_grader
Installing collected packages: wrapt, widgetsnbextension, pyee, pfzy, json5,
jedi, comm, async-lru, tiktoken, playwright, InquirerPy, python-on-whales,
ipywidgets, fica, ipylab, otter_grader, jupyterlab-server, jupyter-lsp,
jupyterlab, jupyter
  Attempting uninstall: wrapt
    Found existing installation: wrapt 2.0.0
    Uninstalling wrapt-2.0.0:
      Successfully uninstalled wrapt-2.0.0
  Attempting uninstall: widgetsnbextension
    Found existing installation: widgetsnbextension 3.6.10
    Uninstalling widgetsnbextension-3.6.10:
      Successfully uninstalled widgetsnbextension-3.6.10
  Attempting uninstall: tiktoken
    Found existing installation: tiktoken 0.12.0
    Uninstalling tiktoken-0.12.0:
      Successfully uninstalled tiktoken-0.12.0
  Attempting uninstall: ipywidgets
    Found existing installation: ipywidgets 7.7.1
    Uninstalling ipywidgets-7.7.1:
      Successfully uninstalled ipywidgets-7.7.1
Successfully installed InquirerPy-0.3.4 async-lru-2.0.5 comm-0.2.3 fica-0.4.1
ipylab-1.1.0 ipywidgets-8.1.8 jedi-0.19.2 json5-0.12.1 jupyter-1.1.1 jupyter-
lsp-2.3.0 jupyterlab-4.4.10 jupyterlab-server-2.28.0 otter_grader-6.1.3
pfzy-0.3.4 playwright-1.55.0 pyee-13.0.0 python-on-whales-0.79.0 tiktoken-0.11.0
widgetsnbextension-4.0.15 wrapt-1.17.3
Collecting kaleido
  Downloading kaleido-1.1.0-py3-none-any.whl.metadata (5.6 kB)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
(5.24.1)
Collecting plotly
  Downloading plotly-6.3.1-py3-none-any.whl.metadata (8.5 kB)
Collecting choreographer>=1.0.10 (from kaleido)
  Downloading choreographer-1.2.0-py3-none-any.whl.metadata (6.8 kB)
Collecting logistro>=1.0.8 (from kaleido)
  Downloading logistro-2.0.1-py3-none-any.whl.metadata (3.9 kB)
Requirement already satisfied: orjson>=3.10.15 in
/usr/local/lib/python3.12/dist-packages (from kaleido) (3.11.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from kaleido) (25.0)

```



```

Collecting pytest-timeout>=2.4.0 (from kaleido)
  Downloading pytest_timeout-2.4.0-py3-none-any.whl.metadata (20 kB)
Requirement already satisfied: narwhals>=1.15.1 in
/usr/local/lib/python3.12/dist-packages (from plotly) (2.10.0)
Requirement already satisfied: simplejson>=3.19.3 in
/usr/local/lib/python3.12/dist-packages (from choreographer>=1.0.10->kaleido)
(3.20.2)
Requirement already satisfied: pytest>=7.0.0 in /usr/local/lib/python3.12/dist-
packages (from pytest-timeout>=2.4.0->kaleido) (8.4.2)
Requirement already satisfied: iniconfig>=1 in /usr/local/lib/python3.12/dist-
packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (2.3.0)
Requirement already satisfied: pluggy<2,>=1.5 in /usr/local/lib/python3.12/dist-
packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in
/usr/local/lib/python3.12/dist-packages (from pytest>=7.0.0->pytest-
timeout>=2.4.0->kaleido) (2.19.2)
Downloading kaleido-1.1.0-py3-none-any.whl (66 kB)
      66.3/66.3 kB
2.9 MB/s eta 0:00:00
Downloading plotly-6.3.1-py3-none-any.whl (9.8 MB)
      9.8/9.8 MB
76.1 MB/s eta 0:00:00
Downloading choreographer-1.2.0-py3-none-any.whl (56 kB)
      56.4/56.4 kB
4.6 MB/s eta 0:00:00
Downloading logistro-2.0.1-py3-none-any.whl (8.6 kB)
Downloading pytest_timeout-2.4.0-py3-none-any.whl (14 kB)
Installing collected packages: plotly, logistro, pytest-timeout, choreographer,
kaleido
  Attempting uninstall: plotly
    Found existing installation: plotly 5.24.1
    Uninstalling plotly-5.24.1:
      Successfully uninstalled plotly-5.24.1
Successfully installed choreographer-1.2.0 kaleido-1.1.0 logistro-2.0.1
plotly-6.3.1 pytest-timeout-2.4.0

```

```

[2]: # Initialize Otter
import otter
grader = otter.Notebook("hw3.ipynb")

```

## Homework 03 – Optimizers and Backpropagation

CS 189, Fall 2025

Welcome to Homework 3! In this assignment, you will use your newfound skills in backpropagation to implement an autodifferentiation library from scratch. You'll extend the single-variable autograd that we learned in lecture to function with general tensors, mimicking how `torch`'s autograd is implemented in research and industry.

---

## 0.1 Due Date: November 7th, 2025

This assignment is due on **November 7th, 2025**. You must submit your work to Gradescope by this deadline. Please refer to the syllabus for the [Slip Day policy](#). No late submissions will be accepted beyond the details outlined in the Slip Day policy.

### 0.1.1 Submission Tips

- **Plan ahead:** We strongly encourage you to submit your work several hours before the deadline. This will give you ample time to address any submission issues.
- **Reach out for help early:** If you encounter difficulties, contact course staff well before the deadline. While we are happy to assist with submission issues, we cannot guarantee responses to last-minute requests.

### 0.1.2 Key Learning Objectives

1. Learn how backpropagation is implemented in libraries like PyTorch
2. Understand how optimizers are implemented, and the advantages of Adam

---

### 0.1.3 Collaboration Policy

You are encouraged to discuss high-level concepts with your peers. However: - All submitted work must be written in your own words and code. - Do not share or copy solutions directly. - List any collaborators (students you worked with) in the line below. Include their name and SID:

**Your Collaborators: TODO**

### 0.1.4 AI Tools Usage Disclosure

We allow the use of AI tools (e.g., ChatGPT, Copilot) **only as support**, not as a replacement for your own reasoning. To ensure transparency, you must acknowledge any use of AI tools.

Please answer with one of the following options: - **A) I did not use any AI tools for this homework.** - **B) I used AI tools in the following way(s):**

(describe briefly, e.g., “Used ChatGPT to get hints for debugging a NumPy indexing error”)

**Your Answer: TODO**

---

### 0.1.5 Grading Breakdown

| Question | Manual Grading? | Points |
|----------|-----------------|--------|
| q1       | No              | 10     |
| q2       | No              | 20     |
| q3       | No              | 10     |
| q4       | No              | 10     |

| Question     | Manual Grading? | Points    |
|--------------|-----------------|-----------|
| <b>Total</b> |                 | <b>50</b> |

```
[3]: from __future__ import annotations
import numpy as np
from dataclasses import dataclass
import matplotlib.pyplot as plt
from collections import deque
!pip install typst
import typst
import re
np.random.seed(189)
```

Collecting typst

Downloading

typst-0.14.1-cp38-abi3-manylinux\_2\_17\_x86\_64.manylinux2014\_x86\_64.whl.metadata  
(2.6 kB)

Downloading

typst-0.14.1-cp38-abi3-manylinux\_2\_17\_x86\_64.manylinux2014\_x86\_64.whl (25.6 MB)  
25.6/25.6 MB

65.9 MB/s eta 0:00:00

Installing collected packages: typst

Successfully installed typst-0.14.1

## Introducing BearTensor

Before we can dig into our multivariable calculus, we first need a good abstraction for tensors, which are just  $n$ -dimensional arrays (similar to `numpy`'s `ndarray`). `numpy` doesn't provide autodifferentiation for its `ndarray`, so we can build a wrapper around it. For this homework assignment we are going to be using the `BearTensor` class, which you can think of as a custom implementation of `torch.Tensor` that acts as a wrapper around `numpy` arrays.

Take a look at the `BearTensor`, `BearGrad`, and `BearParent` classes below.

### Question 1: Building the Computation Graph

#### Conceptual Overview:

Our model needs some way to build a computation graph for backpropagation so that it knows where to pass down gradients during the backwards pass. The way that PyTorch does this, is that every time you combine two values to compute a third one in the forward pass, it will keep track of this internally. We want to implement a similar functionality for `BearTensor`; everytime we add, subtract, divide, etc. two `BearTensor` to get a third one, we should keep track of the fact that the third tensor's **parents** are the two source tensors.

Here is an example of what a computation graph might look like from lecture:

#### Your Task:

Currently, our `BearTensor` has no operations. Implement the following functions in the class: - Addition (`__add__`) - Subtraction (`__sub__`) - Multiplication (`__mul__`) - Power (`__pow__`) - Matrix

multiplication (`__matmul__`) - Dot product (`dot`) - Sum (`sum`) - Mean (`mean`) - ReLU (`relu`) - Sigmoid (`sigmoid`)

A few things to remember: 1. You should return a new `BearTensor` from these functions without modifying the existing one 2. Remember to set the parents of this new Tensor to be the two source tensors; this is how we will build a computation graph 3. Remember to set the correct gradient function as the first argument of `BearParent.BearGrad`; this will tell us how to compute the downstream gradient during the backward pass

Additionally, we are going to be doing a lot NumPy operations in this Homework. It is important that you think about how the shapes of different arrays will change due to NumPy broadcasting rules, and reshape when needed. Some examples of this that are relevant to this HW are: - NumPy will automatically broadcast 1-element arrays to a scalar (ex. when computing the dot product of two vectors) - NumPy will broadcast array of shape  $(N, 1)$  to just  $(N,)$  (ex. when a matrix times a matrix gives a vector)

**Note:** The public tests that we've provided in this notebook are fully-comprehensive and often very simple. You should build more complex graphs and manually verify that your computed gradients and values are correct. Our hidden tests are more comprehensive and will check for full correctness

```
[20]: # @title
def _unbroadcast(grad, target_shape):
    # if numpy collapses dimensions during op  $(N,1) * (N,N) = (N,)$ , add
    # trailing dimensions
    # e.g
    # vector result (3,)
    # original tensor (3,1)
    # convert [1,2,3] to [[1],[2],[3]]
    while grad.ndim < len(target_shape):
        grad = np.expand_dims(grad, axis=-1)

    # if numpy stretches dimensions during op  $(N,) + (N,N) = (N,N)$ , collapse
    # extra dimensions
    # e.g
    # vector result = (3,3)
    # original tensor = (1,)
    # sum over rows to get (3,)
    while grad.ndim > len(target_shape):
        grad = grad.sum(axis=0)

    # Sum across broadcasted axes
    for axis, size in enumerate(target_shape):
        if size == 1:
            grad = grad.sum(axis=axis, keepdims=True)

    # 4. Reshape exactly
    return grad.reshape(target_shape)

@dataclass
```

```

class BearGrad:
    '''Stores how to compute the downstream gradient from the upstream gradient
    - `op_str` is just a string describing the operation; you don't need to use
    ↪ this for this HW, but it may help in debugging if you print out what
    ↪ operations create your computation graph.
    - `fn` is a function that takes in the upstream loss gradient and outputs
    ↪ the loss gradient that should be passed downstream. This essentially applies
    ↪ the chain rule at the current node in the computation graph.
    '''
    fn: callable
    op_str: str | None = None

@dataclass
class BearParent:
    '''This class represents a parent of a node; a node must track its parents
    ↪ so that it knows who to propagate its gradients to.
    - `grad` is the `BearGrad` object for this parent; we can apply the `fn`
    ↪ from this to compute the gradient that should be passed downstream.
    - `parent` is the `BearTensor` object for the parent
    '''
    parent: BearTensor
    grad: BearGrad

    def __hash__(self):
        return id(self)

    def __eq__(self, other):
        return self is other

class BearTensor:
    '''`BearTensor`: This represents a node in our computation graph
    - `value` is the underlying data in the Tensor; this is computed during the
    ↪ forward pass
    - `parents` keeps track of the parents in the computation graph to whom we
    ↪ pass our gradient to
    - `adjoint` is the gradient (the transpose of it technically) that we
    ↪ compute in the backward pass
    '''
    def __init__(self, name: str, value: np.ndarray, parents: list[BearParent]
    ↪ | None = None):
        self.name = name
        self.value = value
        self.parents = parents if parents is not None else []
        self.adjoint: float | np.ndarray = 0.0

    def __add__(self, other: BearTensor) -> BearTensor:

```

```

        if self.value.shape != other.value.shape:
            raise ValueError("Shapes must match")
        result_value = self.value + other.value
        s_shape = self.value.shape
        o_shape = other.value.shape
        return BearTensor(f"{self.name} + {other.name}",
                           result_value,
                           [
                               BearParent(self, BearGrad(lambda g, s=s_shape:
↪_unbroadcast(g, s))),
                               BearParent(other, BearGrad(lambda g, s=o_shape:
↪_unbroadcast(g, s))),
                           ],
                           )

    def __sub__(self, other: BearTensor) -> BearTensor:
        if self.value.shape != other.value.shape:
            raise ValueError("Shapes must match")
        result_value = self.value - other.value
        s_shape = self.value.shape
        o_shape = other.value.shape
        return BearTensor(f"{self.name} - {other.name}",
                           result_value,
                           [
                               BearParent(self, BearGrad(lambda g, s=s_shape:
↪_unbroadcast(g, s))),
                               BearParent(other, BearGrad(lambda g, s=o_shape:
↪_unbroadcast(g, s))),
                           ]
                           )

    def __mul__(self, other: BearTensor) -> BearTensor:
        if self.value.shape != other.value.shape:
            raise ValueError("Shapes must match")
        result_value = self.value * other.value
        s_shape = self.value.shape
        o_shape = other.value.shape
        return BearTensor(f"{self.name} * {other.name}",
                           result_value,
                           [
                               BearParent(self, BearGrad(lambda g,
↪s=s_shape,y=other.value: _unbroadcast(g*y, s))),
                               BearParent(other, BearGrad(lambda g,
↪s=o_shape,x=self.value: _unbroadcast(g*x, s))),
                           ]
                           )

```

```

def __pow__(self, power: float) -> BearTensor:
    result_value = self.value ** power
    s_shape = self.value.shape
    return BearTensor(f"{self.name} ** {power}",
                      result_value,
                      [
                          BearParent(self, BearGrad(lambda g, x=self.
↪value, p=power, s=s_shape:_unbroadcast(g * (p * (x ** (p - 1))), s))),
                      ]
                      )

def __matmul__(self, other: BearTensor) -> BearTensor:
    # This may be helpful to avoid shape mismatch errors
    def ensure_2d(x):
        '''If x is a scalar or 1-D, convert to 2-D'''
        x = np.asarray(x)
        if x.ndim == 0:          # scalar
            return x.reshape(1, 1)
        elif x.ndim == 1:       # vector
            return x.reshape(-1, 1) # column vector convention
        else:                   # already 2D
            return x
    result_value = self.value @ other.value
    s_shape = self.value.shape
    o_shape = other.value.shape
    return BearTensor(
        f"{self.name} @ {other.name}",
        result_value,
        [
            # dL/dX = dL/dZ @ Y^T
            BearParent(self, BearGrad(lambda g, Y=other.value, s=s_shape:
↪_unbroadcast(g @ Y.T, s))),

            # dL/dY = X^T @ dL/dZ
            BearParent(other, BearGrad(lambda g, X=self.value, s=o_shape:
↪_unbroadcast(X.T @ g, s))),
        ],
    )

def dot(self, other: BearTensor) -> BearTensor:
    if self.value.ndim != 1 or other.value.ndim != 1:
        raise ValueError("dot() only supports 1-D BearTensors (like torch.
↪dot).")
    result_value = np.dot(self.value, other.value)
    # required to reshape by autograder
    result_value = result_value.reshape((1,))

```

```

s_shape = self.value.shape
o_shape = other.value.shape
return BearTensor(
    f"dot({self.name}, {other.name})",
    result_value,
    [
        #  $dL/dx = g * y$ 
        BearParent(self, BearGrad(lambda g, y=other.value, s=s_shape:
                                   _unbroadcast(g * y, s))),
        #  $dL/dy = g * x$ 
        BearParent(other, BearGrad(lambda g, x=self.value, s=o_shape:
                                   _unbroadcast(g * x, s))),
    ],
)

def sum(self) -> BearTensor:
    result_value = np.sum(self.value)
    # required to reshape by autograder
    result_value = result_value.reshape((1,))
    return BearTensor(
        f"sum({self.name})",
        result_value,
        [
            BearParent(self, BearGrad(lambda g, x=self.value:
                                       np.ones_like(x) * g)),
        ],
    )

def sum_without_collapse(self):
    result_value = np.sum(self.value, axis=1, keepdims=True)
    return BearTensor(
        f"sum({self.name})",
        result_value,
        [
            BearParent(self, BearGrad(lambda g, x=self.value:
                                       np.ones_like(x) * g)),
        ],
    )

def mean(self) -> BearTensor:
    result_value = np.mean(self.value) # shape ()
    D = self.value.shape[1] if self.value.ndim >= 2 else self.value.size
    invN = 1.0 / D
    return BearTensor(
        f"mean({self.name})",
        result_value,
        [

```



```

        BearParent(self, BearGrad(lambda g, x=self.value, c=invN:
                                   np.ones_like(x) * (g * c))),
    ],
)

def mean_without_collapse(self) -> BearTensor:
    result_value = np.mean(self.value, axis=1, keepdims=True)
    D = self.value.shape[1] if self.value.ndim >= 2 else self.value.size
    invN = 1.0 / D
    return BearTensor(
        f"mean({self.name})",
        result_value,
        [
            BearParent(self, BearGrad(lambda g, x=self.value, c=invN:
                                       np.ones_like(x) * (g * c))),
        ],
    )

def relu(self) -> BearTensor:
    result_value = np.maximum(self.value, 0)
    mask = (self.value > 0).astype(self.value.dtype) # local derivative
    return BearTensor(
        f"ReLU({self.name})",
        result_value,
        [
            BearParent(self, BearGrad(lambda g, m=mask: g * m)),
        ],
    )

def sigmoid(self) -> BearTensor:
    # Forward
    sig = 1.0 / (1.0 + np.exp(-self.value))
    return BearTensor(
        f"sigmoid({self.name})",
        sig,
        [
            BearParent(self, BearGrad(lambda g, s=sig: g * s * (1.0 - s))),
        ],
    )

```

**Disclaimer:** The Q1 test cases will NOT test that your adjoint calculations are correct until you implement Q2; if you are struggling with the next question, we recommend also looking back at your solution to Q1.

```
[21]: grader.check("q1")
```

```
[21]: q1 results: All test cases passed!
```

Below, we have provided some code that will allow you to visualize your computation graphs and an example for how to use it. You are not required to use nor understand how this works, but it may help you debug while working on this homework.

```
[22]: # @title
def to_typst_math(s: str) -> str:
    processed_s = s
    processed_s = processed_s.replace(" @ ", " ")
    processed_s = re.sub(r"([a-zA-Z])([0-9]+)", r"\1_2", processed_s)
    processed_s = re.sub(r"\.([a-zA-Z]+)", r'."\1"', processed_s)
    processed_s = re.sub(r"([a-zA-Z]+)_([a-zA-Z_]+)", r'\1_"2"', processed_s)
    processed_s = re.sub(r"\b(relu|sum)\b", r'"\1"', processed_s)
    return processed_s

def _traverse_graph(
    root: BearTensor,
) -> tuple[list[BearTensor], list[tuple[BearTensor, BearTensor, str]]]:
    nodes, edges = set(), set()
    visited = set()

    def build(v):
        if v not in visited:
            visited.add(v)
            nodes.add(v)
            for p in v.parents:
                edges.add((p.parent, v, p.grad.op_str))
                build(p.parent)

    build(root)
    return list(nodes), list(edges)

def _minimize_crossings(
    nodes_by_level: list[list[BearTensor]],
    edges: list[tuple[BearTensor, BearTensor, str]],
    iterations: int = 4,
) -> list[list[BearTensor]]:
    node_to_parents = {n: [] for level in nodes_by_level for n in level}
    node_to_children = {n: [] for level in nodes_by_level for n in level}
    for p, c, _ in edges:
        if p in node_to_children and c in node_to_parents:
            node_to_children[p].append(c)
            node_to_parents[c].append(p)

    ordered_levels = [list(level) for level in nodes_by_level]
```

```

for _ in range(iterations):
    for level_idx in range(1, len(ordered_levels)):
        child_level = ordered_levels[level_idx - 1]
        child_ranks = {node: rank for rank, node in enumerate(child_level)}

        def barycenter_down(node):
            children = node_to_children.get(node, [])
            if not children:
                return -1
            avg_rank = sum(child_ranks.get(c, 0) for c in children) /
↪len(children)
            return avg_rank

        ordered_levels[level_idx].sort(key=barycenter_down)

    for level_idx in range(len(ordered_levels) - 2, -1, -1):
        parent_level = ordered_levels[level_idx + 1]
        parent_ranks = {node: rank for rank, node in
↪enumerate(parent_level)}

        def barycenter_up(node):
            parents = node_to_parents.get(node, [])
            if not parents:
                return -1
            avg_rank = sum(parent_ranks.get(p, 0) for p in parents) /
↪len(parents)
            return avg_rank

        ordered_levels[level_idx].sort(key=barycenter_up)

    return ordered_levels

def _assign_node_levels(
    root: BearTensor,
    nodes: list[BearTensor],
    edges: list[tuple[BearTensor, BearTensor, str]],
) -> dict[BearTensor, tuple[float, float]]:
    node_to_id = {node: i for i, node in enumerate(nodes)}
    id_to_node = {i: node for i, node in enumerate(nodes)}

    adj = {node_to_id[n]: [] for n in nodes}
    for p, c, _ in edges:
        if c in node_to_id and p in node_to_id:
            adj[node_to_id[c]].append(node_to_id[p])

    levels = {n: -1 for n in nodes}

```

```

if root in levels:
    levels[root] = 0

q = deque([root])
max_level = 0
visited_bfs = {root}

while q:
    u = q.popleft()
    u_id = node_to_id[u]
    for v_id in adj.get(u_id, []):
        v = id_to_node.get(v_id)
        if v and v not in visited_bfs:
            levels[v] = levels[u] + 1
            max_level = max(max_level, levels[v])
            visited_bfs.add(v)
            q.append(v)

nodes_by_level = [[] for _ in range(max_level + 1)]
for node, level in levels.items():
    if level != -1:
        nodes_by_level[level].append(node)

for level_nodes in nodes_by_level:
    level_nodes.sort(key=lambda n: n.name)

nodes_by_level = _minimize_crossings(nodes_by_level, edges)

node_coords = {}
for level, nodes_in_level in enumerate(nodes_by_level):
    num_in_level = len(nodes_in_level)
    for rank, node in enumerate(nodes_in_level):
        y_pos = rank - (num_in_level - 1) / 2.0
        x_pos = -level
        node_coords[node] = (x_pos, y_pos)

return node_coords

def _generate_typst_source(
    nodes: list[BearTensor],
    edges: list[tuple[BearTensor, BearTensor, str]],
    coords: dict[BearTensor, tuple[float, float]],
) -> str:
    node_to_id = {node: i for i, node in enumerate(nodes)}

    header = """#import "@preview/fletcher:0.5.8": diagram, node, edge

```

```

#set page(width: auto, height: auto, margin: 10mm, fill: white)
#set text(font: "Linux Libertine", size: 10pt)

#diagram(cell-size: (40mm, 40mm), {
"""

    node_definitions = ""
    for n in nodes:
        if n not in coords:
            continue
        x, y = coords[n]
        node_id = node_to_id[n]
        math_name = to_typst_math(n.name)
        shape_str = str(n.value.shape)
        content = f""block(stroke: 0.5pt, inset: 8pt, radius: 4pt,
↪[\n#align(center, ${math_name}$)\n#line(length: 100%)\n#text(size: 9pt,
↪`shape: {shape_str}`)\n])""
        node_definitions += f'    node(({x}, {y}), {content}, name:
↪"{node_id}")\n'

    edge_definitions = ""
    LABEL_LENGTH_THRESHOLD = 20

    for parent, child, op_str in edges:
        from_id = node_to_id.get(parent)
        to_id = node_to_id.get(child)
        if from_id is None or to_id is None:
            continue

        op_str = op_str or ""
        math_op_str = to_typst_math(op_str)
        label = f"${math_op_str}$"

        if len(op_str) > LABEL_LENGTH_THRESHOLD:
            edge_definitions += f'    edge(label("{from_id}"), label("{to_id}"),
↪"->", label: {label}, label-sep: 4em)\n'
        else:
            edge_definitions += (
                f'    edge(label("{from_id}"), label("{to_id}"), "->", label:
↪{label})\n'
            )

    footer = "})"

    return header + node_definitions + edge_definitions + footer

```

```

def draw_graph(root: BearTensor, output_typ_path: str = "computational_graph.
↳typ"):
    nodes, edges = _traverse_graph(root)
    coords = _assign_node_levels(root, nodes, edges)
    typst_source = _generate_typst_source(nodes, edges, coords)

    with open(output_typ_path, "w", encoding="utf-8") as f:
        f.write(typst_source)
    typst.compile(output_typ_path, output=output_typ_path.replace(".typ", ".
↳pdf"))

# Example code
a = BearTensor("a", np.array([2, 3]))
b = BearTensor("b", np.array([1, 1]))
c = a + b
draw_graph(c, "demo_graph.typ")

```

## Question 2:

After solving the previous question, your `BearTensor` should be able to construct the computation graph from the forward pass. In this part, we will implement the backwards pass, using the `BearGrad` functionality from above to propagate the gradients downstream.

## Conceptual Overview

When training networks, we want to backpropagate the gradients starting from the very final node, which produced our training loss. To do this efficiently, we adopt a recursive approach; the loss node propagates its gradient back to its parent nodes. Then, every parent node propagates its gradient one layer further back; we keep recursively doing this until gradients have been propagated across the whole computation graph.

In lecture, we implemented single-variable `autograd` with two passes when `.backward()` is called: a “reset” pass that sets up counters for each node in our computational graph, and then a recursive backward pass to iterate through the entire graph and accumulate each node’s adjoint.

We can actually perform both of these steps at once using an algorithm called topological sort. This addresses the core problem we are trying to solve: a node can only propagate its gradient backward after it has received gradients from all its children.

[Topological sort](#) starts from the root node (where we call `.backward()`), and traverses our computational graph to determine a correct order of operations. From Wikipedia: it provides a linear ordering that ensures that for any edge  $u \rightarrow v$  in our computational graph,  $u$  comes before  $v$  in the ordering. It turns out that *depth-first search (DFS)* actually gives us a topological sort!

You should first implement topological sort, then backprop; this ensures that for deep networks we do not encounter a stack overflow error. One implementation of this is [Kahn’s algorithm](#) which is iterative instead of recursive, avoiding stack overflow errors for deep networks. Alternatively, you can use the implementation discussed in lecture.

## Your Task:

Fill out `topological_sort` first, then `reset_children`, and `backward` below to implement back-propagation. After completing this, you finally have your very own PyTorch implementation!

```
[7]: def topological_sort(node):  
    '''Return a list of nodes ordered topologically'''  
    visited = set()  
    sorted_nodes = []  
    def DFS(node, visited, sorted_nodes):  
        if node in visited: return  
        visited.add(node)  
        for p in node.parents:  
            DFS(p.parent, visited, sorted_nodes)  
        sorted_nodes.append(node)  
        return sorted_nodes  
    return DFS(node, visited, sorted_nodes)
```

```
[24]: def reset_children(self):  
    """Resets the gradient in the current node to zero and all nodes before it  
    ↪ in the computation graph."""  
    unresetted = deque()  
    self.adjoint = 0.0  
    unresetted.append(self)  
    while unresetted:  
        node = unresetted.popleft()  
        for p in node.parents:  
            p.parent.adjoint = 0.0  
            unresetted.append(p.parent)  
    pass  
  
def backward(self):  
    from collections import defaultdict  
    """  
    Take a node in the computation graph, reset all gradients, and perform  
    ↪ backpropagation  
    to compute the adjoints (gradients) for all nodes in the graph.  
  
    Hint: After resetting the gradients, what should the gradient at the  
    ↪ current node be?  
    """  
    reset_children(self)  
    sorted_nodes = topological_sort(self)  
    sorted_nodes.reverse()  
  
    self.adjoint = 1.0  
    for node in sorted_nodes:  
        for p in node.parents:  
            downstream = p.grad.fn(node.adjoint)
```

```
p.parent.adjoint = p.parent.adjoint + downstream
```

```
BearTensor.reset_children = reset_children  
BearTensor.backward = backward
```

```
[9]: grader.check("q2")
```

[9]: q2 results: All test cases passed!

### Question 3: Optimizers

Now that we are able to compute the gradients for all the parameters in our model, we want to be able to optimize them. In lecture we talked about Gradient Descent; in this homework, we will implement two widely-used optimizers, Stochastic Gradient Descent and Adam. It is highly recommended that you complete the paper/written portion of the homework before attempting this question.

#### Your Task:

Implement the SGD, Momentum, and Adam optimizers.

```
[10]: class Optimizer:  
    def __init__(self, params: list[BearTensor], lr: float):  
        self.params = params  
        self.lr = lr  
  
    def zero_grad(self):  
        for p in self.params:  
            p.adjoint = np.zeros_like(p.value)  
  
    def step(self):  
        raise NotImplementedError  
  
class SGD(Optimizer):  
    def __init__(self, params: list[BearTensor], lr: float):  
        super().__init__(params, lr)  
  
    def step(self):  
        nodes = self.params  
        for i in range(len(nodes)):  
            nodes[i].value = nodes[i].value - self.lr * nodes[i].adjoint  
  
class Momentum(Optimizer):  
    def __init__(self, params: list[BearTensor], lr: float, beta: float = 0.9):  
        super().__init__(params, lr)  
        self.beta = beta  
        self.velocities = [np.zeros_like(p.value) for p in self.params]
```



```

    def step(self):
        nodes = self.params
        for i in range(len(nodes)):
            # self.velocities[i] = self.beta * self.velocities[i] + (1-self.
↪beta) * nodes[i].adjoint
            # nodes[i].value = nodes[i].value - self.lr * self.velocities[i]
            self.velocities[i] = self.beta * self.velocities[i] + nodes[i].
↪adjoint
            nodes[i].value -= self.lr * self.velocities[i]

class Adam(Optimizer):
    def __init__(
        self,
        params: list[BearTensor],
        lr: float,
        beta1: float = 0.9,
        beta2: float = 0.999,
        eps: float = 1e-8,
    ):
        super().__init__(params, lr)
        self.beta1 = beta1
        self.beta2 = beta2
        self.eps = eps
        self.t = 0
        self.ms = [np.zeros_like(p.value) for p in self.params]
        self.vs = [np.zeros_like(p.value) for p in self.params]

    def step(self):
        nodes = self.params
        self.t += 1
        for i in range(len(nodes)):
            self.ms[i] = self.beta1 * self.ms[i] + (1-self.beta1) * nodes[i].
↪adjoint
            self.vs[i] = self.beta2 * self.vs[i] + (1-self.beta2) * np.
↪square(nodes[i].adjoint)
            m_hat = self.ms[i] / (1-self.beta1**self.t)
            v_hat = self.vs[i] / (1-self.beta2**self.t)
            nodes[i].value = nodes[i].value - self.lr * m_hat / (np.sqrt(v_hat) +
↪self.eps)
        pass

```

```
[11]: grader.check("q3")
```

```
[11]: q3 results: All test cases passed!
```

The code below compares the convergence rates of all 3 optimizers. Play around with the learning

rates and other hyperparameters to see how the results change.

```
[13]: import numpy as np
import matplotlib.pyplot as plt
from copy import deepcopy

def compare_optimizers(optimizers):
    np.random.seed(0)
    X = BearTensor("X", np.random.randn(1000, 24))
    y = BearTensor("y", np.random.randn(1000, 1))

    n_steps = 100
    results = {}

    W1_init = np.random.randn(24, 12)
    W2_init = np.random.randn(12, 1)

    for opt_class, title, lr in optimizers:
        W1 = BearTensor("W1", deepcopy(W1_init))
        W2 = BearTensor("W2", deepcopy(W2_init))
        params = [W1, W2]

        optimizer = opt_class(params, lr=lr)
        losses = []

        for step in range(n_steps):
            optimizer.zero_grad()

            hidden = (X @ W1).sigmoid()
            output = hidden @ W2
            loss = ((output - y) ** 2).mean()

            loss.backward()
            losses.append(loss.value.item())
            optimizer.step()

        results[title] = losses

    plt.figure(figsize=(8, 5))
    for title, losses in results.items():
        plt.plot(losses, label=title, linewidth=2)
    plt.xlabel("Epoch")
    plt.ylabel("Loss")
    plt.title("Optimizer Convergence Comparison (Same Init)")
    plt.legend()
    plt.grid(True)
    plt.show()
```

```
compare_optimizers([
    (SGD, "SGD", 0.05),
    (Momentum, "Momentum", 0.05),
    (Adam, "Adam", 0.05)
])
```

/tmp/ipython-input-3218941297.py:227: RuntimeWarning:

overflow encountered in exp

/usr/local/lib/python3.12/dist-packages/numpy/\_core/\_methods.py:127:  
RuntimeWarning:

overflow encountered in reduce

/tmp/ipython-input-3218941297.py:106: RuntimeWarning:

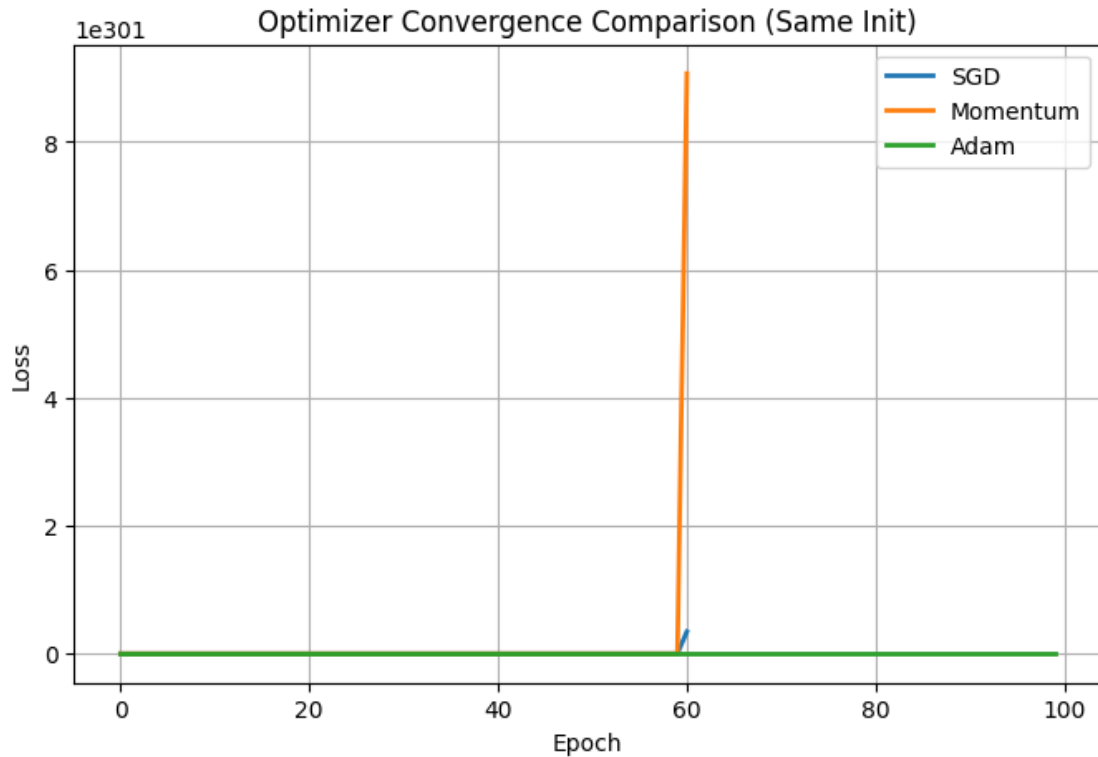
overflow encountered in square

/tmp/ipython-input-3218941297.py:135: RuntimeWarning:

overflow encountered in matmul

/tmp/ipython-input-3218941297.py:232: RuntimeWarning:

invalid value encountered in multiply



#### Question 4: Train a model!

In this section, we're going to use the backpropagation engine and optimizers you created to train a model to predict wine quality scores!

##### Conceptual Overview:

Unfortunately, training our model is going to be slightly more involved than just using PyTorch. Our simple implementation does not allow us to define layers. Instead, we are going to define **BearTensors**, combine them manually to do a forward pass (which will automatically construct a computation graph!) and then finally backpropagate on our loss.

We are going to use the Wine Quality dataset, which contains 1599 samples of red wine. Each sample has 11 features representing various chemical properties, and the task is to predict the wine quality score (between 0 and 10). You can choose any loss function and optimizer (as long as you implemented it!) of your choice, feel free to play around with hyperparameters and different model sizes!

##### Your Tasks:

Train a model to predict wine quality scores. Your model should:

- Use at least one hidden layer
- Use an activation function for hidden layers
- Achieve a Mean Squared Error (MSE) of  $\leq 2.0$  on the dataset

**You need to fill in the `predict()` function that uses your architecture to return a prediction given an input datapoint. The autograder will use this to test your imple-**

mentation.

**Note:** If your model takes more than 30 seconds to train it could crash the autograder; you should easily be able to **achieve a MSE of 2** on the entire dataset with a 1-layer neural net for full credit. The staff solution takes <1 second to train, so if your code is slow, you may be constructing your graph inefficiently.

**You will receive 50% credit for this question if your MSE is 3.0 on the dataset.**

Tips for training models: - Output your training and test loss every epoch; are they smoothly going down? If not, you may need to change the learning rate - Ensure your computation graph is being built properly (remember our visualization helper functions from earlier!) - Changing the number of layers or adding/removing bias terms can be a useful way to control how much your model is overfitting/underfitting

```
[50]: from sklearn.datasets import fetch_openml
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import numpy as np
import matplotlib.pyplot as plt

# DO NOT CHANGE THE PREPROCESSING CODE
data = fetch_openml("wine-quality-red", as_frame=True)
X = data.data.to_numpy()
y = data.target.to_numpy().astype(float)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.4, random_state=42, shuffle=True
)

X_tensor = BearTensor("X_train", X_train) # shape: (N_train, 11)
# input_dim, in this case input_dim = 11
y_tensor = BearTensor("y_train", y_train.reshape(-1,1)) # shape: (N_train,1)

# Add your training code here!

# create test tensors for later evaluation
X_test_tensor = BearTensor("X_test", X_test)
y_test_tensor = BearTensor("y_test", y_test.reshape(-1,1))

# layers
# sum (1599,11) -> (1599,1)
# ReLU
# mean (1599,1) -> (1599,1)
# ReLU
```

```

# sigmoid (1599,1) -> (1599,1)

# initialise weights using He initialisation
def he_init_tensor(name: str, in_features: int, out_features: int) -> BearTensor:
    std = np.sqrt(2.0 / in_features)
    W_value = np.random.randn(in_features, out_features) * std
    return BearTensor(name, W_value)

def zeros_bias(name: str, out_features: int):
    return BearTensor(name, np.zeros((1, out_features)))

W1 = he_init_tensor("W1", 1, 1)
b1 = zeros_bias("b1", 1)
W2 = he_init_tensor("W2", 1, 1)
b2 = zeros_bias("b2", 1)
W3 = he_init_tensor("W3", 1, 1)
b3 = zeros_bias("b3", 1)
params = [W1, b1, W2, b2, W3, b3]
# --- Prediction function ---

def predict(x):
    """Output the scalar prediction for a single training datapoint x"""
    N = x.value.shape[0] # batch size
    b_shaper = BearTensor("b_shaper", np.ones((N, 1)))
    z = (x.sum_without_collapse() @ W1 + (b_shaper @ b1)).relu()
    z = (z.mean_without_collapse() @ W2 + (b_shaper @ b2)).relu()
    z = (z @ W3 + (b_shaper @ b3)).sigmoid() # sigmoid outputs 0 to 1
    scaleTen = BearTensor("scaleTen", np.full((N, 1), 10.0))
    z = z*scaleTen # wine score is between 0 and 10
    return z

def train_step(X_tensor: BearTensor, y_tensor: BearTensor, epochs, optim = None):
    if optim is None:
        optim = Adam(params, lr=1e-3)

    for epoch in range(epochs):
        y_pred = predict(X_tensor)
        loss = ((y_pred - y_tensor) ** 2).mean()
        loss.backward()
        optim.step()
        X_tensor.reset_children()
        MSE = float(loss.value)

```

```

        if epoch % 100 == 0:
            print(f"Epoch {epoch+1}/{epochs}, MSE={MSE:.4f}")

def evaluate(X_test_tensor: BearTensor, y_test_tensor: BearTensor):
    y_hat = predict(X_test_tensor)

    mse_ten = ((y_hat - y_test_tensor) ** 2).mean()

    mse = float(mse_ten.value)

    y_true = y_test_tensor.value
    y_pred = y_hat.value

    X_test_tensor.reset_children()

    print(f"Test MSE: {mse:.4f}")

train_step(X_tensor, y_tensor, 1000)
evaluate(X_test_tensor, y_test_tensor)

```

```

Epoch 1/1000, MSE=8.6530
Epoch 101/1000, MSE=8.3411
Epoch 201/1000, MSE=8.0635
Epoch 301/1000, MSE=7.7426
Epoch 401/1000, MSE=7.3349
Epoch 501/1000, MSE=6.7966
Epoch 601/1000, MSE=5.8959
Epoch 701/1000, MSE=3.7187
Epoch 801/1000, MSE=0.8421
Epoch 901/1000, MSE=0.6533
Test MSE: 0.6503

```

## 0.2 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a zip file for you to submit. **Please save before exporting!**

```

[51]: ## Use this cell if you are running the notebook in Google Colab to install the
      ↪ necessary dependencies, this may take a few minutes
      if IS_COLAB:
          !apt-get install -y texlive texlive-xetex pandoc

```

```

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.

```

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono  
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java  
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12  
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0  
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13  
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet  
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils  
teckit tex-common tex-gyre texlive-base texlive-binaries  
texlive-fonts-recommended texlive-latex-base texlive-latex-extra  
texlive-latex-recommended texlive-pictures texlive-plain-generic tipa  
xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java  
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java  
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho  
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai  
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv  
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec  
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments  
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl  
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex  
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex  
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono  
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java  
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12  
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0  
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13  
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet  
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils  
teckit tex-common tex-gyre texlive texlive-base texlive-binaries  
texlive-fonts-recommended texlive-latex-base texlive-latex-extra  
texlive-latex-recommended texlive-pictures texlive-plain-generic  
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all



20200910-1 [6,367 kB]  
Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common  
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]  
Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64  
1.38-4ubuntu1 [60.0 kB]  
Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64  
0.35-15build2 [16.5 kB]  
Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64  
0.19-3build2 [64.7 kB]  
Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64  
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]  
Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6  
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]  
Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64  
1.0.2-1build4 [45.2 kB]  
Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64  
2.13.1-1 [1,221 kB]  
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all  
2.004.5-6.1 [4,532 kB]  
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all  
20201225-1build1 [397 kB]  
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all  
20180621-3.1 [10.2 MB]  
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java  
all 18-1 [4,720 B]  
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-  
java all 43-1 [10.8 kB]  
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-  
java all 1.2-2 [60.3 kB]  
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1  
amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]  
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration  
all 1.18 [5,336 B]  
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64  
3.0.2-7ubuntu2.11 [50.1 kB]  
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems  
all 3.3.5-2ubuntu1.2 [228 kB]  
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1  
[5,100 B]  
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7  
kB]  
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all  
0.1.1-2 [12.6 kB]  
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick  
all 1.7.0-3ubuntu0.2 [52.5 kB]  
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all  
0.3.2-1ubuntu0.1 [24.9 kB]  
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0

amd64 3.0.2-7ubuntu2.11 [5,114 kB]  
 Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsyntax2  
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]  
 Get:31 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64  
 2.5.11+ds1-1 [421 kB]  
 Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53  
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]  
 Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2  
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]  
 Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzip-0-13 amd64  
 0.13.72+dfsg.1-1.1 [27.0 kB]  
 Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all  
 1:1.0.5-0ubuntu2 [578 kB]  
 Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64  
 1:7.7+6build2 [94.6 kB]  
 Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all  
 2.004.5-6.1 [9,471 kB]  
 Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style  
 all 12.2-1ubuntu1 [185 kB]  
 Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64  
 1.41-4build2 [61.3 kB]  
 Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64  
 2.5.11+ds1-1 [699 kB]  
 Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all  
 20180621-3.1 [6,209 kB]  
 Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-  
 binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]  
 Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all  
 2021.20220204-1 [21.0 MB]  
 Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-  
 recommended all 2021.20220204-1 [4,972 kB]  
 Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base  
 all 2021.20220204-1 [1,128 kB]  
 Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-  
 recommended all 2021.20220204-1 [14.4 MB]  
 Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all  
 2021.20220204-1 [14.3 kB]  
 Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all  
 1:1.8.16-2 [207 kB]  
 Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all  
 1:1.8.16-2 [5,199 kB]  
 Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures  
 all 2021.20220204-1 [8,720 kB]  
 Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra  
 all 2021.20220204-1 [13.9 MB]  
 Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-  
 generic all 2021.20220204-1 [27.5 MB]  
 Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21

```

[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 5s (34.4 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 125080 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...

```

```

Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...

```

```

Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack .../33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.

```

```

Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...

```

```

Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...

```

```

Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.

```



```
Running mktexlsr /var/lib/texmf ... done.  
Building format(s) --all.  
This may take some time... done.
```

```
[52]: # Save your notebook first, then run this cell to export your submission.  
grader.export(pdf=False, run_tests=True)
```

Running your submission against local test cases...

Your submission received the following results when run against available test cases:

q1 results: All test cases passed!

q2 results: All test cases passed!

q3 results: All test cases passed!

<IPython.core.display.HTML object>